

Daily Geometry Drill #2

Sheet by Henry Koduthore

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Circle Equations (centre $(-g, -f)$, $r = \sqrt{g^2 + f^2 - c}$) · Tangents · Line–Circle Tests · Chord Lengths.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Complete the square, read the centre and radius, then write or test circle data on sight.

- Write down the centre of the circle $x^2 + y^2 - 6x - 8y + 24 = 0$.
- Write down the radius of the circle $x^2 + y^2 - 6x - 8y + 24 = 0$.
- Write down the equation of the circle with centre $(2, -1)$ and radius 5.
- Write down the radius of the circle $x^2 + y^2 - 4x + 6y - 12 = 0$.
- The point $(x, 4)$ lies on the circle $x^2 + y^2 = 25$ with $x > 0$. Find x .
- The radius from the origin to the point $(3, 4)$ on $x^2 + y^2 = 25$ has what gradient?
- The tangent to $x^2 + y^2 = 25$ at the point $(3, 4)$ has what gradient?
- Write down the equation of the circle with centre $(-1, 2)$ and radius 3.
- The centre of $x^2 + y^2 - 6x - 8y + 24 = 0$ is $(3, 4)$. Find the perpendicular distance of this centre from the line $4x + 3y - 9 = 0$.
- A chord of the circle $x^2 + y^2 = 25$ is at perpendicular distance 3 from the origin. Find the length of the chord.

Section B Manipulation Drills

(≤50 s each)

Combine the new circle tools with last drill's line machinery. Complete the square first, then test.

- Which of the following is the equation of the circle with centre $(3, -2)$ and radius 4?
 - $x^2 + y^2 - 6x + 4y - 3 = 0$
 - $x^2 + y^2 - 6x + 4y + 9 = 0$
 - $x^2 + y^2 + 6x - 4y - 3 = 0$

- D) $x^2 + y^2 - 6x - 4y - 3 = 0$
2. The circle $x^2 + y^2 + 2x - 4y - 11 = 0$ has what radius?
- A) 2
B) 3
C) 4
D) $\sqrt{5}$
3. The point $(1, 2)$ lies on the circle $x^2 + y^2 = 5$. The equation of the tangent to the circle at $(1, 2)$ is?
- A) $y = -\frac{1}{2}x + \frac{5}{2}$
B) $y = 2x$
C) $y = -2x + 5$
D) $y = \frac{1}{2}x$
4. A circle of form $x^2 + y^2 + 2gx + 2fy + c = 0$ passes through the origin. Deduce c .
- A) $c = 1$
B) $c = 0$
C) $c = g + f$
D) $c = -1$
5. The perpendicular distance from the centre of $x^2 + y^2 = 25$ to the line $y = 2x + 3$ is?
- A) $\frac{3}{\sqrt{5}}$
B) $\frac{3}{5}$
C) 3
D) 2
6. Which of the following is a tangent to the circle $x^2 + y^2 = 4$?
- A) $y = x + 2\sqrt{2}$
B) $y = x + 4$
C) $y = 2x + 2$
D) $y = 2x + 4$
7. A chord of the circle $x^2 + y^2 = 13$ has length $4\sqrt{3}$. What is the perpendicular distance of the chord from the centre?
- A) 3
B) 2

- C) $\sqrt{5}$
D) 1
8. The line $y = mx + 4$ is a tangent to the circle $x^2 + y^2 = 4$. Find the two possible values of m .
A) ± 1
B) $\pm\sqrt{3}$
C) ± 2
D) $\pm\sqrt{2}$
9. The circle with centre $(6, a)$ and radius 6 is tangent to the y -axis and passes through the point $(0, 8)$. Find a .
A) 1
B) 3
C) 5
D) 8
10. Write down the length of the chord cut from the line $y = x + 1$ by the circle $x^2 + y^2 = 25$.

Section C Substitution & Structure(≤ 90 s each)

The numbers were chosen so a clean route exists. Never substitute into the circle — use the distance test.

1. The line $y = 2x + 5$ is a tangent to the circle $x^2 + y^2 = r^2$. Find r^2 .
A) 1
B) 5
C) 25
D) $5\sqrt{5}$
2. The point $(2, -1)$ lies on the circle $x^2 + y^2 + 2gx + 2fy - 3 = 0$ with $g = f$. Find g .
A) -1
B) 0
C) 1
D) 2
3. The circle $x^2 + y^2 - 4x + 2y - 11 = 0$ and the line $x - y + 1 = 0$ have what relationship?
A) The line cuts the circle in two points (secant)
B) The line is a tangent
C) The line misses the circle
D) The line passes through the centre

4. The tangent to $x^2 + y^2 = 9$ at $(2, \sqrt{5})$ has equation?
- A) $2x + \sqrt{5}y = 9$
 - B) $2x - \sqrt{5}y = 9$
 - C) $2x + \sqrt{5}y = 3$
 - D) $\sqrt{5}x + 2y = 9$
5. The midpoint of a chord of $x^2 + y^2 = 40$ is at distance 2 from the centre. Find the chord length.
- A) 12
 - B) $6\sqrt{2}$
 - C) 8
 - D) $4\sqrt{10}$
6. Which of these circles is tangent to the x -axis?
- A) $x^2 + y^2 - 6y = 0$
 - B) $x^2 + y^2 + 4x = 0$
 - C) $x^2 + y^2 - 6x + 9 = 0$
 - D) $x^2 + y^2 - 2y - 3 = 0$
7. The circle $x^2 + y^2 - 8x - 4y - 21 = 0$ cuts the y -axis at two points. The distance between these two points is?
- A) 4
 - B) 6
 - C) 8
 - D) 10
8. The line $y = kx + 5$ touches the circle $x^2 + y^2 - 6x - 8y + 24 = 0$. How many distinct values of k achieve this?
- A) 0
 - B) 1
 - C) 2
 - D) 3

Section D Challenge

(TMUA / SMC difficulty)

Two or three skills at once: distance tests, chord geometry, and tangent-at-point, then check against the options.

1. The line $3x + 4y - 12 = 0$ cuts the circle $x^2 + y^2 - 8x - 6y + 16 = 0$ in two points. Find the length of the chord.
- A) $\frac{18}{5}$

- B) $\frac{12}{5}$
- C) 6
- D) $\frac{36}{5}$

2. A circle with centre $(3, k)$ is tangent to the x -axis and passes through the point $(0, 4)$. Find the radius.

- A) $\frac{25}{8}$
- B) $\frac{25}{9}$
- C) 3
- D) 4

3. The tangent to $x^2 + y^2 = 25$ at the point P (in the first quadrant) passes through the point $A = (7, 1)$. Find the x -coordinate of P .

- A) 2
- B) 3
- C) 4
- D) 5

4. A circle has centre $(1, 4)$ and is tangent to the line $y = 2x + 1$. Find its radius.

- A) $\frac{1}{\sqrt{5}}$
- B) $\frac{1}{5}$
- C) 1
- D) $\sqrt{5}$

5. The circle through the three points $(0, 0)$, $(6, 0)$ and $(3, -1)$ has what centre?

- A) $(3, 4)$
- B) $(3, 2)$
- C) $(4, 3)$
- D) $(5, 0)$

“Tangency is a distance statement in disguise: compare d to r , never solve the intersection.”

Found a mistake or want to contribute a sheet?
Visit the open-source project at github.com/The-CerealDev/speed-maths